

PANAVERSITY • LECTURE 7 • PROJECT 4

Organize The Mess. Safely.

📁 File Organization

🔍 Duplicate Detection

🛡️ Zero Deletions

✅ Human-Approved

A complete record of how a real, messy Panaversity assignment folder was inspected, planned, approved, and reorganized by type — without a single file being deleted, overwritten, or touched without permission.

Prepared for **Syeda Hafsa Fatima Naqvi** | Assignment: Project 4 — Organize The Mess | July 2026

What "Clean" Means Here

Before any code touched a single file, the messy folder was zipped up as a backup copy and worked on separately — the original files on the computer were never opened directly by the script.

1 Sort everything by file type

Word docs, PDFs, HTML files, Jupyter notebooks, CSVs, and images should each land in their own clearly-named folder instead of being scattered across "lecture 2", "lecture 4", "lecture 6" and "lecture 7" assignment folders.

2 Find duplicate files

Some files had been uploaded twice — once inside the messy zip, and once again as a separate loose file — with different names. "Same content, different name" needed to count as a duplicate, not just "same name".

3 Flag anything over 100 MB

Large files are usually the first thing worth reviewing in a messy folder (old videos, big exports, etc). None were expected here, but the check still had to run.

4 Never delete anything

The single hard rule for this task: files can be copied, renamed, and reorganized into new folders — but nothing gets deleted, and nothing gets overwritten. Duplicates get flagged and pointed to, not removed.

The Prompts I Gave The AI

Exactly what was typed into Claude, in order. The second prompt is the one that matters most — it's what turned this from "just run a script" into a safety-checked workflow.

PROMPT 1

"My folder is messy. Please organize files by type, detect duplicate files, identify files larger than 100 MB, but don't delete anything."

"Show me your full plan first. Show me every file you plan to move, rename, or delete. Wait for my approval before doing anything."

PROMPT 2 — THE APPROVAL

"flatten everything
change name of double extension
yes make shortcuts"

PROMPT 3

"show me the code you have done for this"

PROMPT 4 — FIX & CONSOLIDATE

"put it all in one place, and fix the mistakes that are in the code, and put it all together"

"Wait for my approval before doing anything" is the one sentence that turned this into a safe assignment instead of a risky one.

What The Mess Actually Looked Like

Real screenshots of the folders as they existed before any organizing happened.

Desktop

KingsoftData

messy_folder

XuanZhu

Zoom

api key new

API KEY

Hacking Tools

messy_folder.zip

OpenCart URL

4/14/2026 3:00 AM

4/13/2026 2:25 AM

7/7/2026 3:35 AM

5/8/2026 3:30 PM

5/22/2026 9:00 AM

5/7/2026 5:27 AM

4/12/2026 3:43 AM

4/12/2026 7:45 AM

7/7/2026 4:00 AM

4/12/2026 3:57 AM

File folder

File folder

File folder

File folder

Text Document

Text Document

Text Document

Compressed (zipp...

Text Document

1 KB

1 KB

1 KB

607 KB

1 KB

Downloads folder — the messy zip (607 KB, circled) sitting next to unrelated text files like API keys and "Hacking Tools" notes.

project-1-money-detective

project-2-whats-my-grade

project-3-books-dont-match

project-4-organize-the-mess

7/6/2026 8:09 AM

7/6/2026 8:43 AM

7/7/2026 3:35 AM

7/7/2026 4:18 AM

File folder

File folder

File folder

File folder

Assignment folder — four separate project folders (money-detective, whats-my-grade, books-dont-match, organize-the-mess) all mixed together.

Task 1 Lab-report.docx.pdf

Task 2 snake-game.html

Task 2 snake-game-report.html

Task_3_Slide-Presentation

Task_3_Slide-Promptingreport.pdf

6/10/2026 7:32 AM

6/7/2026 3:48 AM

6/7/2026 4:21 AM

6/13/2026 5:17 AM

6/13/2026 5:17 AM

DOCX Document

Microsoft Edge H...

Microsoft Edge H...

WPS PDF Docume...

WPS PDF Docume...

21 KB

41 KB

17 KB

247 KB

102 KB

Lecture 2 assignment files — note the doubled extensions: Task 1 Lab-report.docx.pdf, Task 2 snake-game-report.html, Task 3 Slide-Promptingreport.pdf.

panaguide by Mr.AI App

6/15/2026 2:14 AM

Microsoft Edge H...

87 KB

panaguide by Mr.AI App.html — 87 KB, same duplicate pattern.

hafsa-portfolio

6/23/2026 4:33 AM

Microsoft Edge H...

14 KB

hafsa-portfolio.html — 14 KB, uploaded loose as well as inside the zip.

Project 4 — Organize The Mess

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How The Dry Run & Approval Actually Worked

No file was moved, renamed, or touched until this back-and-forth was complete.

① Full plan shown first

Before touching anything, every file was listed with its current location, its proposed new location, and its new type-folder — in a table, in the chat, in full. Nothing was summarized or hidden.

② Duplicates flagged, not deleted

All 7 duplicate files were listed by name with the exact file they matched — and it was stated clearly that they would only be flagged, never removed, unless told otherwise.

③ Large-file check reported as empty

The >100 MB check was run and reported explicitly as "none found" — an empty result is still a result that needs to be shown, not skipped.

④ Explicit open questions asked back

Two genuine judgment calls were handed back for a decision instead of being assumed: "flatten everything or keep it lecture-wise?" and "clean up the doubled extensions or leave names as-is?" Both were answered directly in the approval message.

⑤ Execution only after the reply

Only once "flatten everything / change name of double extension / yes make shortcuts" was received did any copy, rename, or zip command actually run.

What `organize_files.sh` Actually Does

One bash script, six clear steps. No step deletes or overwrites anything — every operation either reads, or copies to a brand-new location.

1 Unzip a fresh working copy

The original zip is extracted into a scratch folder that gets wiped and rebuilt every time the script runs. The real uploaded zip file itself is never modified.

2 Measure and fingerprint every file

Every file's size is listed, an MD5 "fingerprint" is generated for every file, and files over 100 MB are searched for. Two files with the same fingerprint are guaranteed to have identical content, even if their names are completely different — that's how the 7 duplicates were actually proven, not guessed.

3 Build empty destination folders

Seven folders are created up front — one per file type. *Bug that was caught and fixed here:* the first version tried to create all seven folders in one shortcut command using curly-brace syntax, which silently failed in this environment and created one wrongly-named folder instead of seven. The fixed version creates each folder with its own explicit command.

4 Copy — never move — each file into place

Every file is **copied** (not moved) into its new type folder. Files with messy double-extensions like `.docx.pdf.docx` or `.html.html` are given one clean, correct extension at this step.

5 Replace duplicates with shortcut notes

For each of the 7 duplicate files, instead of copying the same content a second time, a small text file is created explaining that this is a duplicate and pointing to exactly where the real file lives. Zero storage wasted on repeated content.

6 Zip the result — originals untouched

The entire organized folder is zipped into one downloadable file. At no point in any of the six steps was the original uploaded zip or any original file deleted or edited.

Full File Inventory

TYPE	FOLDER	FILES
Word Documents	Documents_Word/	2
PDF Files	PDFs/	6
Web / HTML Files	Web_HTML/	4
Jupyter Notebooks	Notebooks_Python/	3
Spreadsheets (CSV)	Spreadsheets_CSV/	2
Images (PNG)	Images/	3
Original Archive	Archives/	1 (untouched)

Duplicates Detected (flagged, not deleted)

FILE	MATCHES	SIZE	STATUS
Task_1_Lab-report_docx_pdf.docx	Task 1 Lab-report.docx.pdf.docx	21 KB	duplicate
Task_2_snake-game-report_html.html	Task 2 snake-game-report.html.html	17 KB	duplicate
Task_2_snake-game_html.html	Task 2 snake-game.html.html	41 KB	duplicate
Task_3_Slide-Presentation.pdf	Task_3_Slide-Presentation.pdf	247 KB	duplicate
Task_3_Slide-Promptingreport_pdf.pdf	Task_3_Slide-Promptingreport.pdf.pdf	101 KB	duplicate
hafsa-portfolio.html	hafsa-portfolio.html	14 KB	duplicate
panaguide_by_Mr_AI_App.html	panaguide by Mr.AI App.html	87 KB	duplicate

Large-file check (>100 MB)

none found Largest single file was Task_3_Slide-Presentation.pdf at ~247 KB — nowhere close to the 100 MB threshold.

The Result, Verified

Step 6 of the assignment: spot-check that the reorganized files landed exactly where expected. Screenshot taken directly from the downloaded, unzipped output.

This PC > Downloads > Organized_Files > Organized >

Name	Type	Compressed size	Password ...	Size	Ratio	Date modified
Archives	File folder					7/7/2026 11:09 AM
Documents_Word	File folder					7/7/2026 11:09 AM
Images	File folder					7/7/2026 11:09 AM
Notebooks_Python	File folder					7/7/2026 11:09 AM
PDFs	File folder					7/7/2026 11:09 AM
Spreadsheets_CSV	File folder					7/7/2026 11:09 AM
Web_HTML	File folder					7/7/2026 11:09 AM

Downloads → Organized_Files → Organized — all seven type-folders, created at 11:09 AM, exactly matching the approved plan. No stray files left outside a folder.

EXPECTED OUTCOME — ACHIEVED

What I Actually Own Now

"I found 7 duplicate files wasting about 527 KB of redundant storage, rediscovered 21 real files scattered across four different lecture folders, and organized all of it into 7 clear categories — with the original zip file still sitting untouched inside Archives/."

Kept for reuse

`organize_files.sh` saved as a ready-to-run cleanup script for the next messy folder.

Nothing lost

Every original file — inside the zip and the loose duplicates — still exists exactly where it was uploaded from.